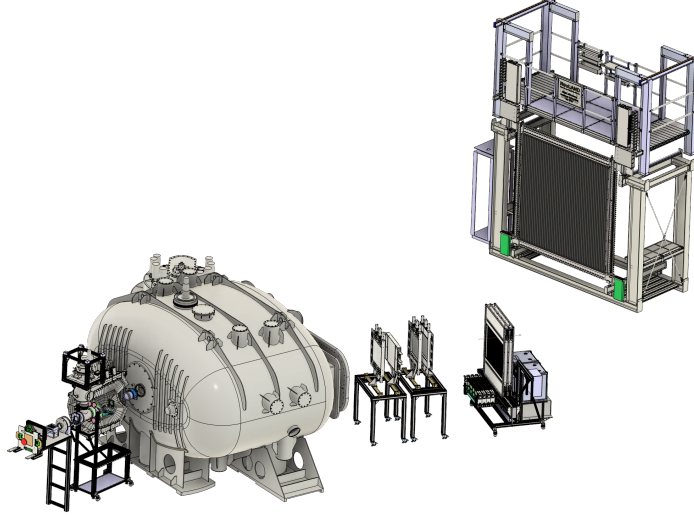

G249 Anlysis Report

Pablo González Rusell



This document is a brief description of the different task performed in the analysis of the G249 experiment. In particular we focus on the work done by the macros of the GitHub repository `g249_analysis`.

1. Calibrations

These macros check the performance of the calibrations of some detectors (FOOTs and NeuLAND for the moment).

1.1. FOOT

For FOOTs, all the calibrations were done in-flight by R3BROOT (in `map2cal` and `cal2hit` classes), so the only correction done after the experiment was charge calibration. The problem is that there is a dependence on the energy with the eta parameter that does not allow to perform a direct conversion between energy and charge (see fig. 1 left).

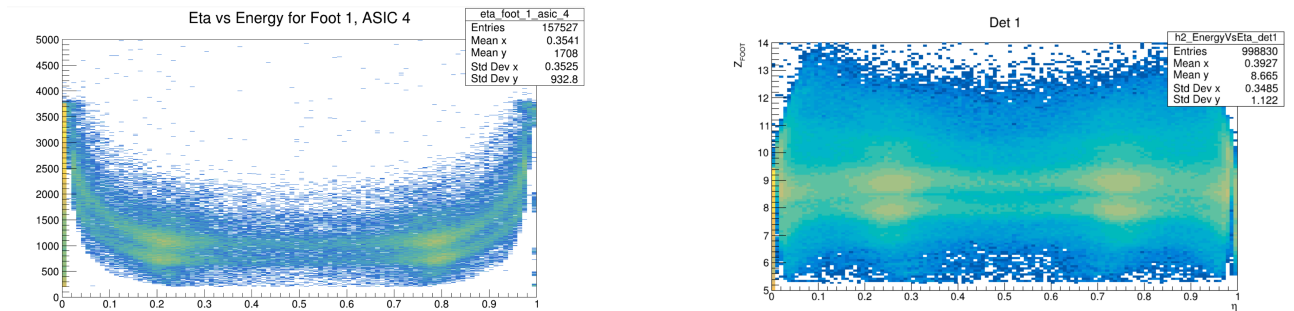


Figure 1: Energy as a function of the eta parameter. Left: With eta correction or charge calibration, only one ASIC. Right: After eta correction and charge calibration for all the ASICs of the detector.

To make the eta correction we perform a high-degree polynomial fit to the $Z = 8$ and $Z = 9$ bands for each ASIC chip and calibrate the energy by making a linear fit (channels, charge) for those bands. This work is done by R3BROOT and, in the macro `check_eta_correction.C`, we plot the charge as a function of eta to show that the former dependence has disappear (see fig. 1 right).

On the other hand, this macro also represent the charge measured in the FOOTs (after all the corrections) against the charge measured in LOS. In fig. 2, we see a clear correlation between both charges. Whereas for charges different from $Z = 8$ and $Z = 9$ we see some saturation, this is not present for our charges of interest.

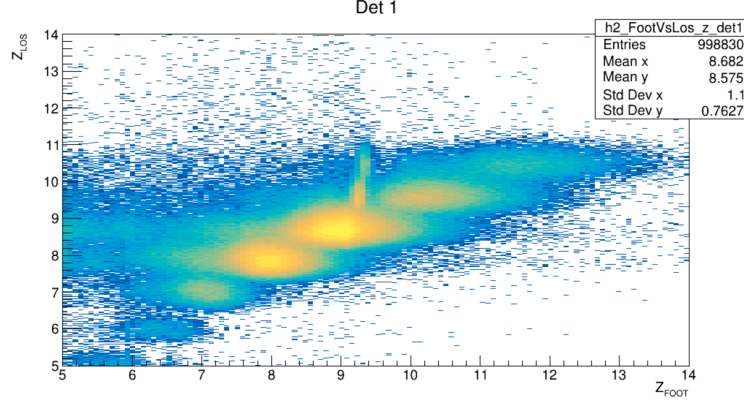


Figura 2: Charge in FOOT vs charge in LOS for the same events

1.2. NeuLAND

2. Simulation

2.1. Neutron emission decay

Here we simulate the decay of the ^{24}O , produced in (p,2p) reaction, by neutron emission. The idea is to use the generator `R3BNeutronGenerator`. There, we will input the mass and charge of the incoming fragment, $A = 24$ and $Z = 8$, for ^{24}O , the kinetic energy per nucleon in the lab frame of the fragment, $\sim 650 \text{ MeV}^{-1}$, the number of emitted neutrons and the energy and width of the resonance that we want to simulate.

The first idea is to reproduce the same resonance measured in [1],

Referencias

- [1] C. R. Hoffman, T. Baumann, D. Bazin, J. Brown, G. Christian, D. H. Denby, P. A. DeYoung, J. E. Finck, N. Frank, J. Hinnefeld, S. Mosby, W. A. Peters, W. F. Rogers, A. Schiller, A. Spyrou, M. J. Scott, S. L. Tabor, M. Thoennessen, and P. Voss. Evidence for a doubly magic ^{24}O . *Physics Letters B*, 672:17–21, 2009.

¹This is the energy of the ^{25}F fragment, in the this approximation we neglect the energy lost by this nucleus in all the materials before the target, and also consider (p,2p) reaction as a quasi elastic reaction, conserving the energy per nucleon.